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CLASS CODE: BSSE2024B

SUBJECT: AI

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PROJECT REPORT 1

PHASE 1 REPORT: SETUP AND SPEECHTOTEXT IMPLEMENTATION

OBJECTIVE

In the objective of Phase 1, I set up environment variables and implement the first phase of the project by converting lecture audio into text using Albased speech recognition model.

TOOLS & TECHNOLOGIES USED:

- **Python:** Main programming language
- **VS Code:** Development environment
- **OpenAI Whisper:** Speech to text model
- **FFmpeg:** Audio processing tool

TASKS PERFORMED:

1. Environment Setup

- Installed Python and configured system environment variables
- Created a project folder for organizing files
- Set up a virtual environment to manage dependencies

2. Library Installation

The following libraries were installed using pip:

- Openai whisper for speech recognition
- torch for running the AI model
- flask (for future web integration)
- transformers (for upcoming NLP tasks)

3. FFmpeg Installation

- Installed FFmpeg to enable audio file processing
- Added FFmpeg to system PATH
- Verified installation using terminal command

4. Speech to Text Implementation

- Created a Python module to handle audio transcription

- Loaded the Whisper pre trained model
- Developed a function to convert audio files into text

Functionality:

- Input: Audio file (.mp3)
- Output: Transcribed text

5. Testing

- Created a test script to verify functionality
- Successfully converted a sample lecture audio file into text
- Debugged issues such as file path errors and missing dependencies

OUTPUT:

The system successfully converted the audio file into text. Screenshots of the output is below:

The screenshot shows a Visual Studio Code editor window titled "AI Lecture Notes Generator". The editor has three tabs open: "speech_to_text.py", "summarizer.py", and "test.py". The "speech_to_text.py" tab is active, displaying the following Python code:

```
1 import whisper
2
3 model = whisper.load_model("base")
4
5 def convert_audio_to_text(audio_path):
6     result = model.transcribe(audio_path)
7     return result["text"]
8
9
```

Below the code editor, the "OUTPUT" panel is visible, showing the output of the program:

```
Output:
What if I told you there was something that you can do right now that would have an immediate positive benefit for your brain, including your mood and your focus? And what if I told you that same thing could actually last a long time and protect your brain from different conditions like depression, Alzheimer's disease or dementia? Would you do it? Yes, I am talking about the powerful effects of physical activity that is simply moving your body, has immediate long lasting and protective benefits for your brain and they can last for the rest of your life. So what I want to do today is tell you a story about how I use my deep understanding of neuroscience as a professor of neuroscience to essentially do an experiment on myself in which I discovered the science underlying why exercise is the most transformative thing that you can do for your brain today. Now as a neuroscientist, I know that our brains, that is the thing in our head right now, what is the most complex structure known to humankind? But it's one thing to talk about the brain and it's another to see it. So here is a real preserved human brain and it's going to illustrate two key areas that we're going to talk about today. The first is the prefrontal cortex right behind your forehead, critical for things like decision-making, focus, attention and your personality. The second key area is located in the temporal lobe shown right here. You have two temporal lobes on your brain, the right and the left and deep in the temporal lobe is a key structure, critical for your ability to form and retain new long-term memories for facts and events and that structure is called the hippocampus. So I've always been fascinated with the hippocampus. How could it be that an event that lasts just a moment, say your first kiss or the moment your first child was born, can form a memory that has changed your brain that lasts an entire lifetime? That's what I want to understand. I wanted to start and record the activity of individual brain cells in the hippocampus as subjects were forming new memories and essentially trying to decode how those brief bursts of electrical activity, which is how neurons communicate with each other, how those brief bursts either allowed us to form a new memory or did not. But a few years ago I did something very unusual in science. As a full professor of neural science, I decided to completely switch my research program because I encountered something that was so amazing with the potential to change so many lives that I had to study it. I discovered and I experienced the brain changing effects of exercise. And I did it in a completely inadvertent way. I was actually at the height of all the memory work that I was doing. Data was pouring in. I was becoming known in my field for all of this memory work. It should have been going great. It was scientifically. But when I stuck my head out of my lab door, I noticed something. I had no social life. I spent too much time listening to those brain cells in a dark room by myself. I didn't move my body at all. I had gained 25 pounds and actually took me many years to realize it was actually miserable and I shouldn't be miserable. I went on a river rafting trip by myself because I had no social life. And I came back thinking, oh my God, I was the weakest person on that trip and I came back with a mission. I said, I'm never going to feel like the weakest person on a river rafting trip again. And that's what made me go to the gym. And I focused my type A personality on going to all the exercise classes at the gym. I tried everything. I went to kickbox, dance, yoga, step class. And at first it was really hard. But what I noticed is that after every sweat-inducing workout that I tried, I had this great mood boost and this great energy boost. And that's what kept me going back to the gym. Well I started feeling stronger
```

The screenshot shows the same Visual Studio Code editor window, but now the "test.py" tab is active. The code in "test.py" is:

```
1 from speech_to_text import convert_audio_to_text
2
3 print("Starting...")
4
5 text = convert_audio_to_text("sample.mp3")
6
7 print("Output:")
8 print(text)
```

The "OUTPUT" panel below shows the output of the test script:

```
Output:
Starting...
What if I told you there was something that you can do right now that would have an immediate positive benefit for your brain, including your mood and your focus? And what if I told you that same thing could actually last a long time and protect your brain from different conditions like depression, Alzheimer's disease or dementia? Would you do it? Yes, I am talking about the powerful effects of physical activity that is simply moving your body, has immediate long lasting and protective benefits for your brain and they can last for the rest of your life. So what I want to do today is tell you a story about how I use my deep understanding of neuroscience as a professor of neuroscience to essentially do an experiment on myself in which I discovered the science underlying why exercise is the most transformative thing that you can do for your brain today. Now as a neuroscientist, I know that our brains, that is the thing in our head right now, what is the most complex structure known to humankind? But it's one thing to talk about the brain and it's another to see it. So here is a real preserved human brain and it's going to illustrate two key areas that we're going to talk about today. The first is the prefrontal cortex right behind your forehead, critical for things like decision-making, focus, attention and your personality. The second key area is located in the temporal lobe shown right here. You have two temporal lobes on your brain, the right and the left and deep in the temporal lobe is a key structure, critical for your ability to form and retain new long-term memories for facts and events and that structure is called the hippocampus. So I've always been fascinated with the hippocampus. How could it be that an event that lasts just a moment, say your first kiss or the moment your first child was born, can form a memory that has changed your brain that lasts an entire lifetime? That's what I want to understand. I wanted to start and record the activity of individual brain cells in the hippocampus as subjects were forming new memories and essentially trying to decode how those brief bursts of electrical activity, which is how neurons communicate with each other, how those brief bursts either allowed us to form a new memory or did not. But a few years ago I did something very unusual in science. As a full professor of neural science, I decided to completely switch my research program because I encountered something that was so amazing with the potential to change so many lives that I had to study it. I discovered and I experienced the brain changing effects of exercise. And I did it in a completely inadvertent way. I was actually at the height of all the memory work that I was doing. Data was pouring in. I was becoming known in my field for all of this memory work. It should have been going great. It was scientifically. But when I stuck my head out of my lab door, I noticed something. I had no social life. I spent too much time listening to those brain cells in a dark room by myself. I didn't move my body at all. I had gained 25 pounds and actually took me many years to realize it was actually miserable and I shouldn't be miserable. I went on a river rafting trip by myself because I had no social life. And I came back thinking, oh my God, I was the weakest person on that trip and I came back with a mission. I said, I'm never going to feel like the weakest person on a river rafting trip again. And that's what made me go to the gym. And I focused my type A personality on going to all the exercise classes at the gym. I tried everything. I went to kickbox, dance, yoga, step class. And at first it was really hard. But what I noticed is that after every sweat-inducing workout that I tried, I had this great mood boost and this great energy boost. And that's what kept me going back to the gym. Well I started feeling stronger
```

CONCLUSION:

Phase 1 was successfully completed by setting up the environment and implementing speech to text functionality using a pretrained AI model.